

Assumptions, diagnostics, and multicollinearity

Data

Data on the number of doctors, hospitals, and beds in 53 randomly selected US counties. Variables include:

- + MDs: number of doctors in the county
- + Hospitals: number of hospitals in the county
- + Beds: number of beds in the hospitals

Goal: Predict the number of doctors using the number of hospitals and the number of beds

Multiple regression model

$$\text{MDs} = \beta_0 + \beta_1 \text{Hospitals} + \beta_2 \text{Beds} + \varepsilon$$

What assumptions does our multiple regression model make?

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What assumptions does our multiple regression model make?

The same assumptions as before:

- + Shape
- + Constant variance
- + Normality
- + Independence
- + Randomness

Shape and constant variance

$$\text{MDs} = \beta_0 + \beta_1 \text{Hospitals} + \beta_2 \text{Beds} + \varepsilon$$

Shape: the shape of the regression model is at least approximately correct

- + In this case, the number of doctors has a linear relationship with the number of hospitals and the number of beds

Constant variance: The variance of the noise term ε is the same for all numbers of hospitals and beds

How do we assess the shape and constant variance assumptions?

Shape and constant variance

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How do we assess the shape and constant variance assumptions?

A residual plot!

Residual plot

$$\widehat{\text{MDs}} = -472.08 + 117.40 \text{ Hospitals} + 1.26 \text{ Beds}$$

What goes on the x and y axes of my residual plot?

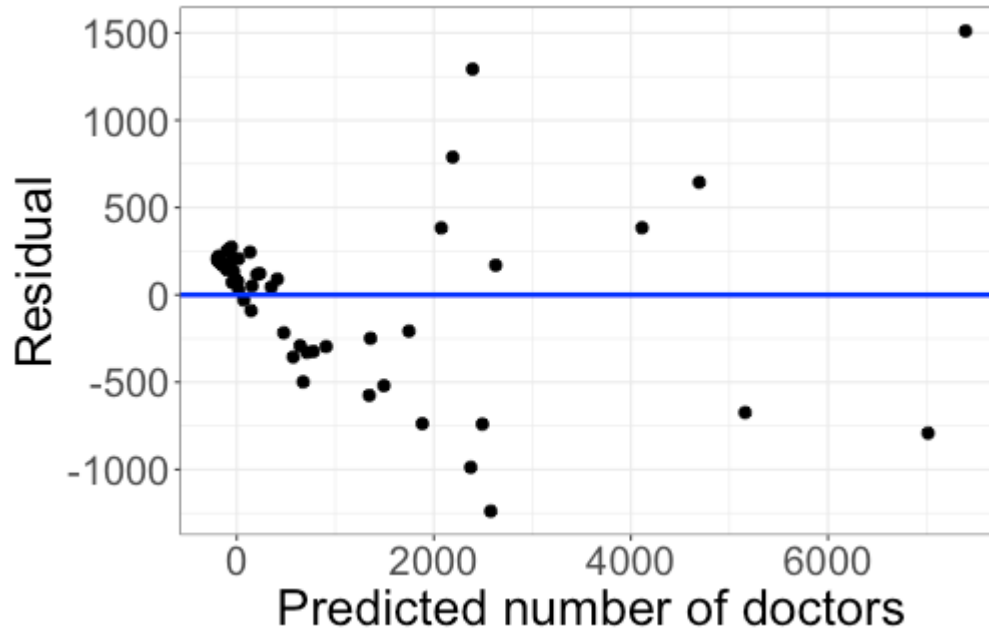
Residual plot

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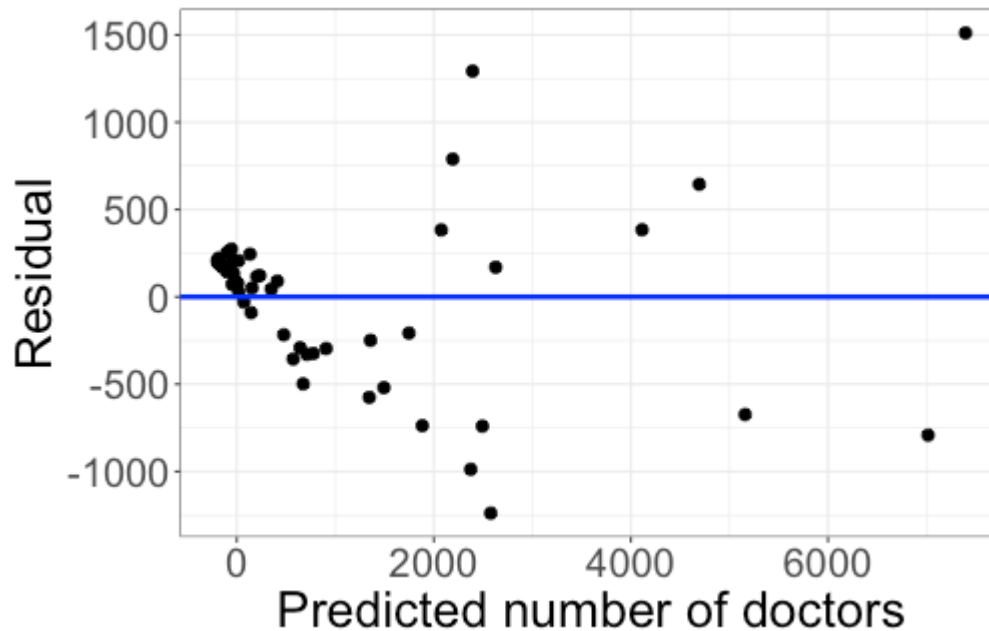
- + x axis: the predicted response ($\widehat{\text{MDs}}$)
- + y axis: the residuals ($\text{MDs} - \widehat{\text{MDs}}$)

Residual plot



How do the shape and constant variance assumptions look?

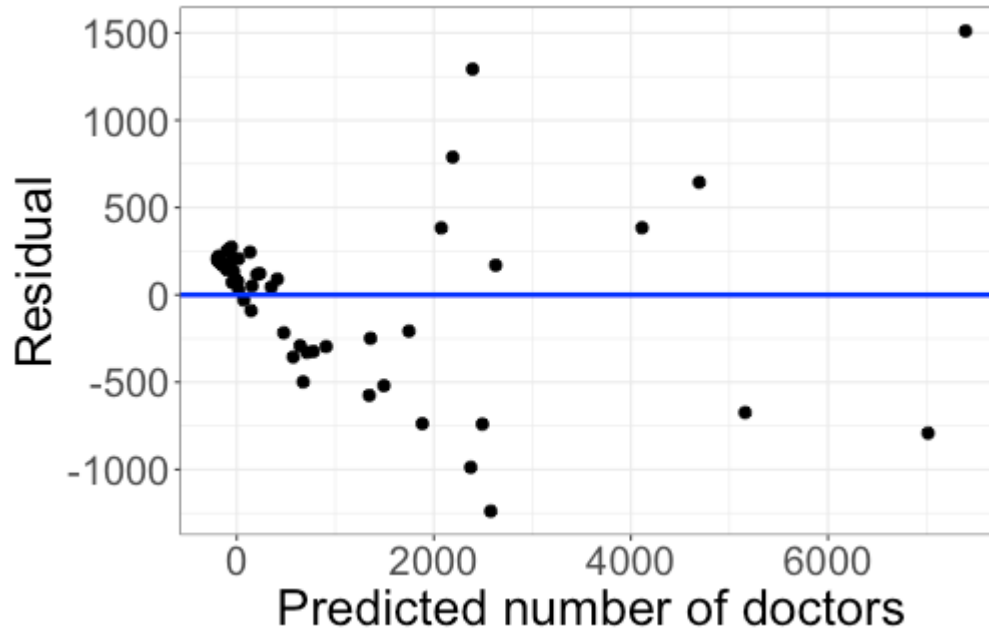
Residual plot



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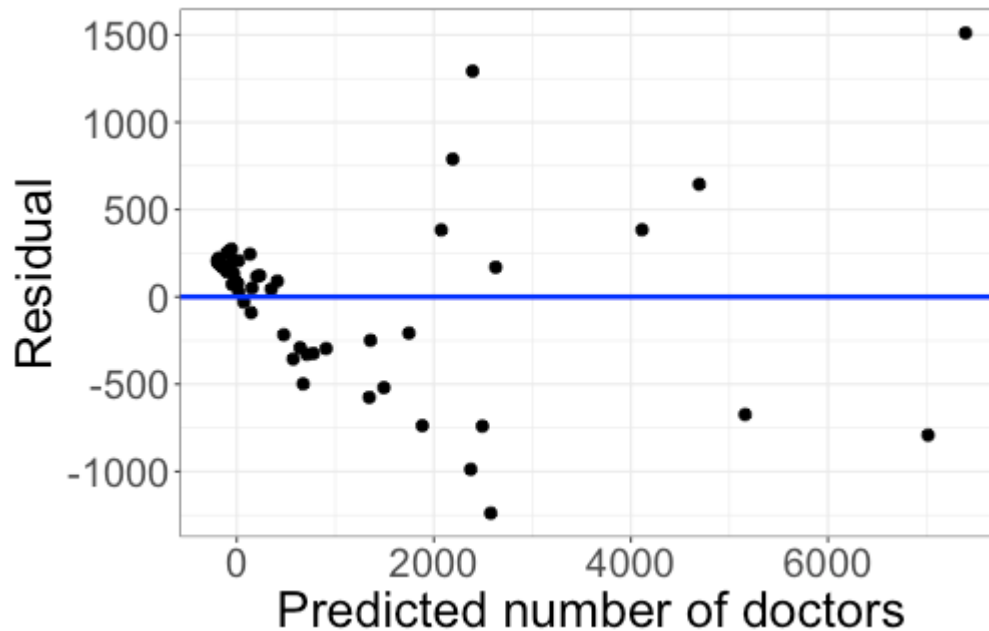
Not good -- violations to both shape and constant variance

Residual plot



How do we address violations of shape and constant variance?

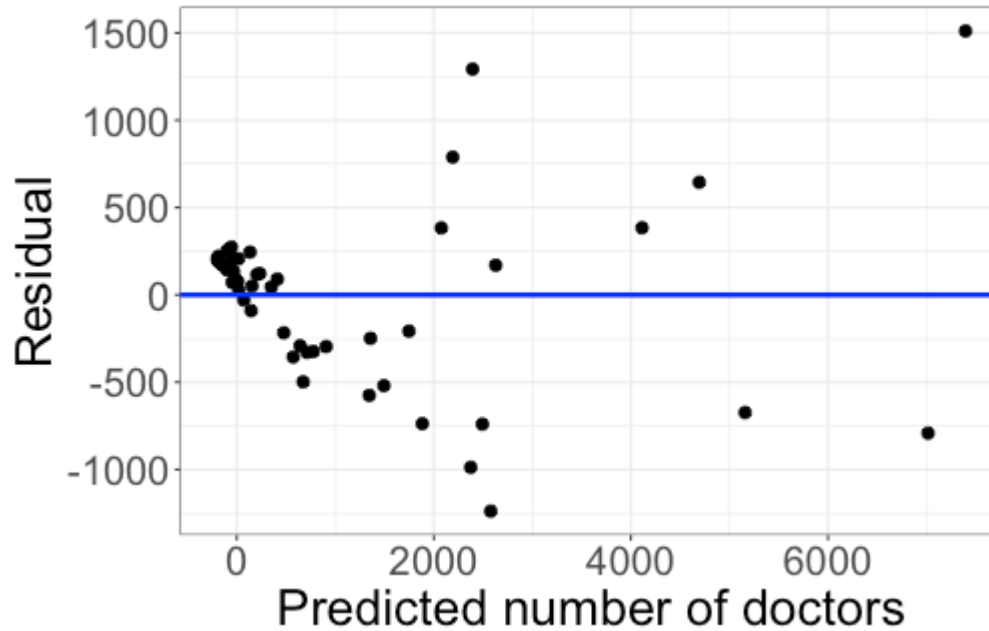
Residual plot



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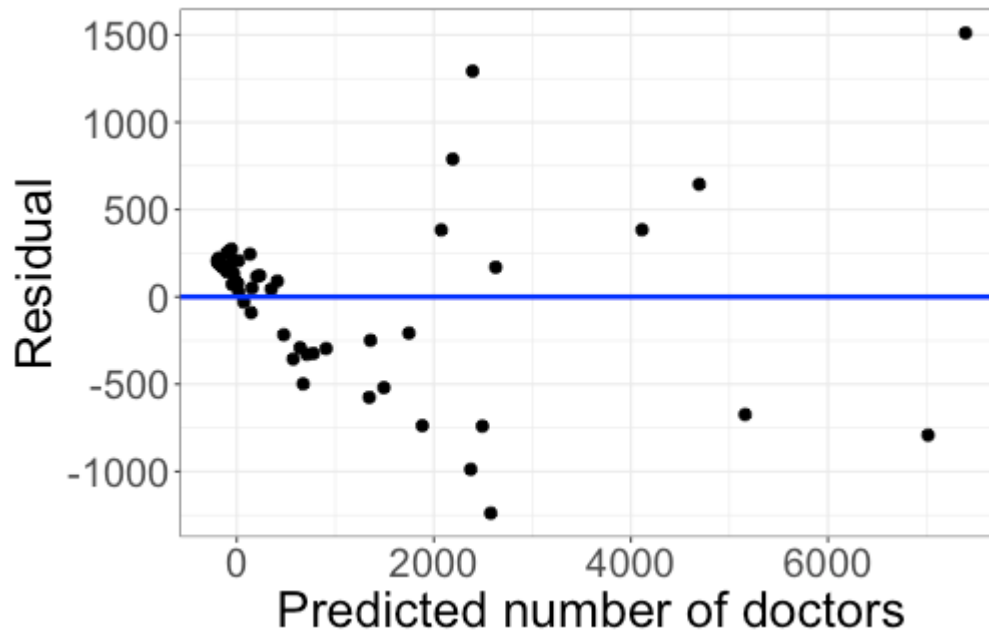
Transformations!

Residual plot



What transformation would you recommend?

Residual plot

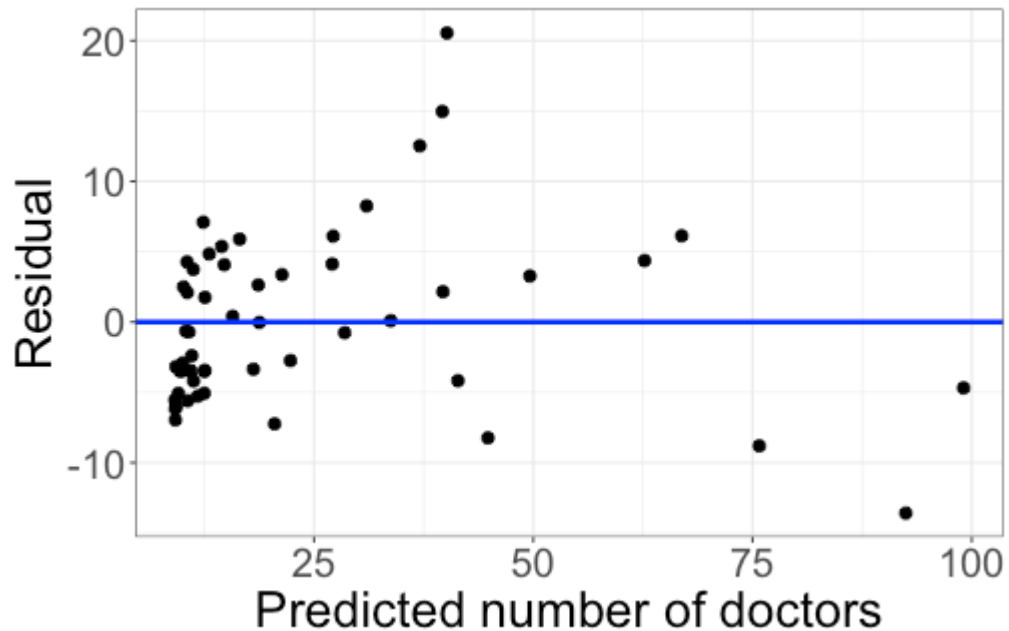


What transformation would you recommend?

Maybe log or square root the response

Transforming the data

$$\sqrt{\widehat{\text{MDs}}} = 3.58 + 2.58 \text{ Hospitals} + 0.012 \text{ Beds}$$



This looks better!

Normality

$$\sqrt{\text{MDs}} = \beta_0 + \beta_1 \text{ Hospitals} + \beta_2 \text{ Beds} + \varepsilon$$

Normality assumption: the noise term ε follows a normal distribution

How do we assess the normality assumption?

Normality

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Normality assumption: the noise term ε follows a normal distribution

How do we assess the normality assumption?

With a QQ plot!

QQ plot

$$\widehat{\sqrt{\text{MDs}}} = 3.58 + 2.58 \text{ Hospitals} + 0.012 \text{ Beds}$$

What goes on the x and y axes of my QQ plot?

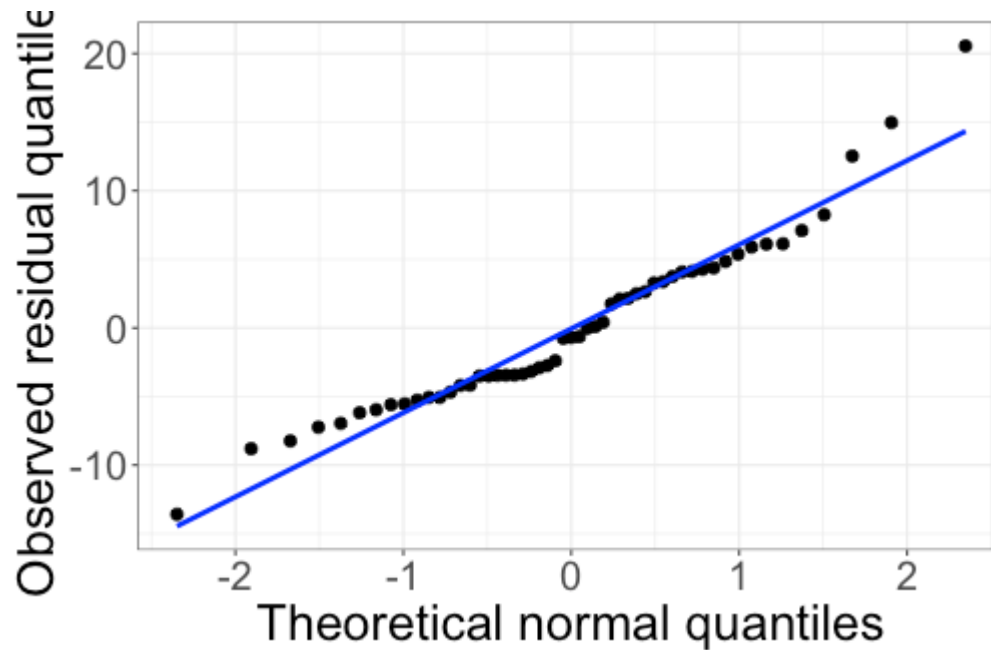
QQ plot

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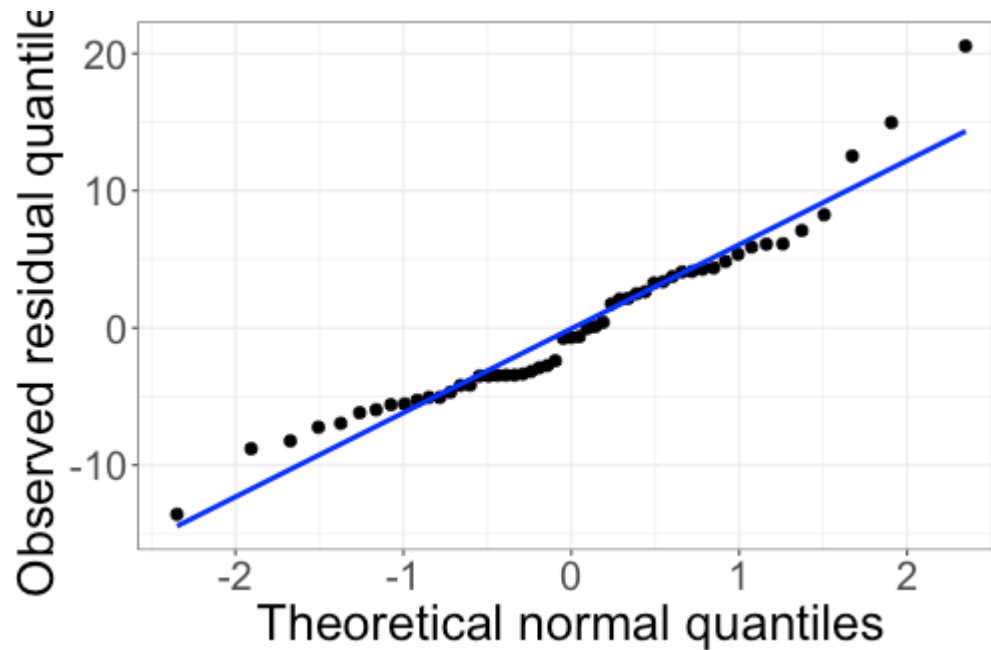
- + x axis: theoretical normal quantiles
- + y axis: observed residual quantiles

QQ plot



How does the QQ plot look?

QQ plot



How does the QQ plot look?

Pretty good!

Independence and randomness

$$\sqrt{\text{MDs}} = \beta_0 + \beta_1 \text{ Hospitals} + \beta_2 \text{ Beds} + \varepsilon$$

Independence assumption: Observations are independent -- one point falling above or below the line has no influence on the location of another point

Randomness assumption: The data are obtained using a random process, such as a random sample or randomized experiment

How do we assess the independence and randomness assumptions?

Independence and randomness

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How do we assess the independence and randomness assumptions?

Think about the data generating process (can't check these assumptions with plots)

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Do independence and randomness seem reasonable for the county health data?

Independence and randomness

Independence assumption: Observations are independent -- one point falling above or below the line has no influence on the location of another point

Randomness assumption: The data are obtained using a random process, such as a random sample or randomized experiment

Do independence and randomness seem reasonable for the county health data?

- + Independence: reasonable to assume the number of hospitals/doctors for one county doesn't affect the number of hospitals/doctors for another county
- + Randomness: we are told the data come from a random sample

Comparing models

With both variables:

$$+ \widehat{\sqrt{\text{MDs}}} = 3.58 + 2.58 \text{ Hospitals} + 0.012 \text{ Beds}$$

$$+ R^2_{adj} = 0.919$$

With just Beds:

$$+ \widehat{\sqrt{\text{MDs}}} = 8.49 + 0.018 \text{ Beds}$$

$$+ R^2_{adj} = 0.899$$

With just Hospitals:

$$+ \widehat{\sqrt{\text{MDs}}} = -2.75 + 6.88 \text{ Hospitals}$$

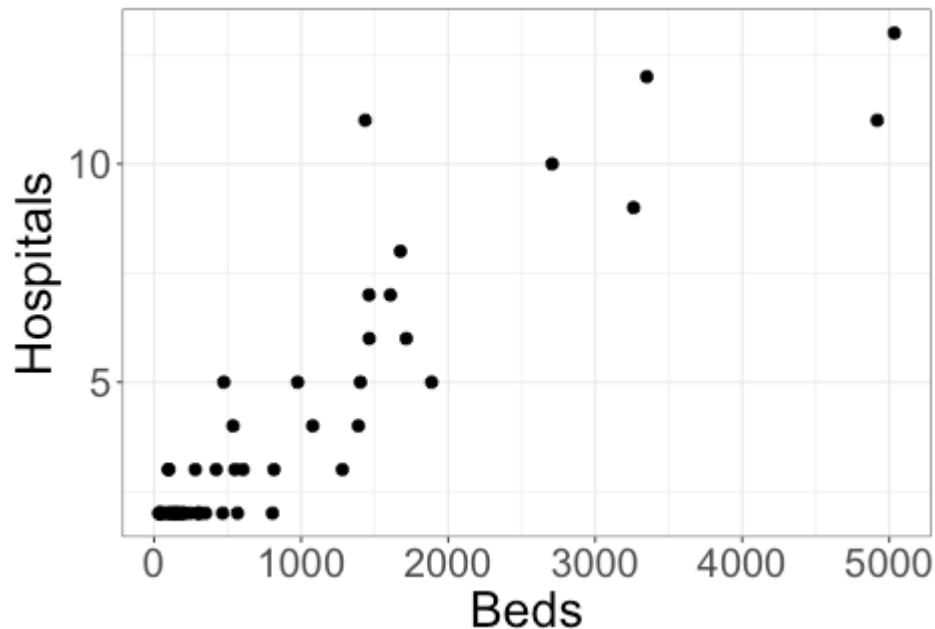
$$+ R^2_{adj} = 0.849$$

Comparing models

Each variable is strongly associated with the response, but using both in the model doesn't improve the R^2_{adj} by much.
Why the small change?

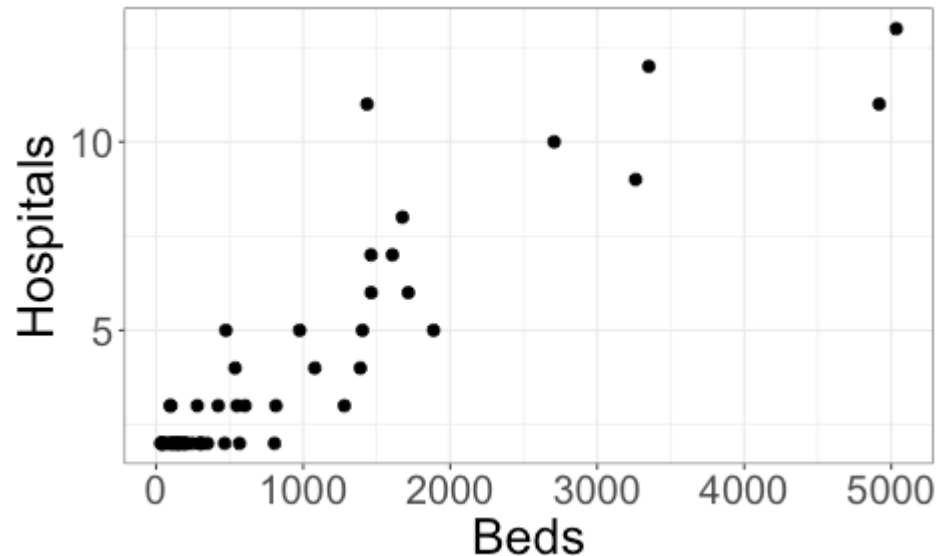
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Multicollinearity

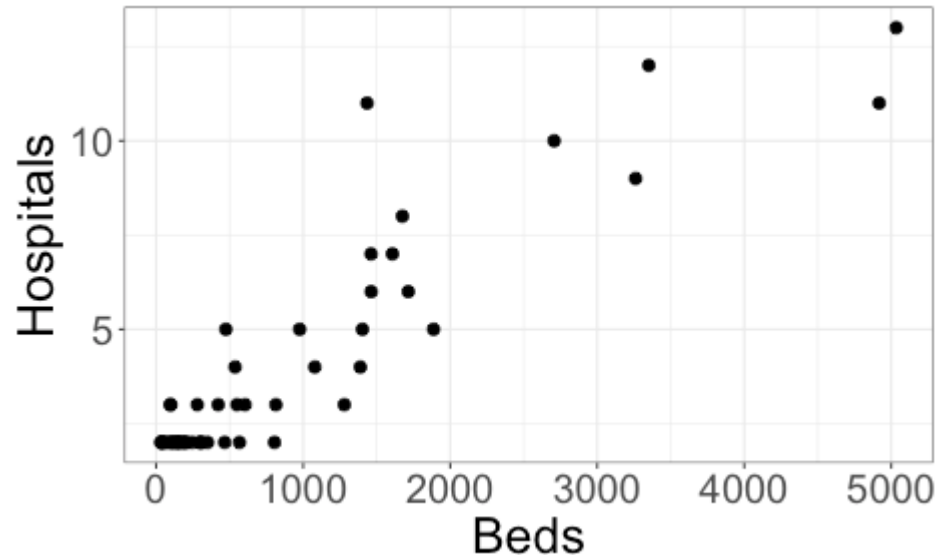
Beds and Hospitals are strongly related, so using both doesn't provide much more information than using just one.



Multicollinearity: one or more predictors is strongly correlated with some combination of the other predictors.

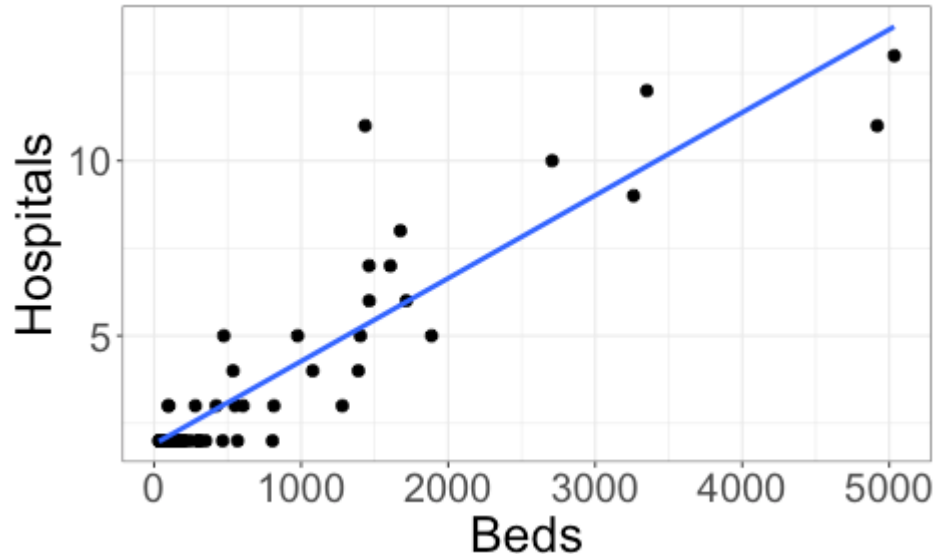
Multicollinearity

Multicollinearity: one or more predictors is strongly correlated with some combination of the other predictors.



$r = 0.91$ (correlation between Beds and Hospitals)

Measuring multicollinearity



$$\widehat{\text{Beds}} = -514.68 + 349.15 \text{ Hospitals}$$

$$R^2 = 0.82 \text{ (for predicting Beds from Hospitals)}$$

$$\text{Variance inflation factor: } VIF = \frac{1}{1 - R^2} = \frac{1}{1 - 0.82} = 5.56$$

Variance inflation factors

Motivation: High multicollinearity means a predictor can be estimated well by some combination of other predictors

Variance inflation factor: Let X_i be a predictor in the model, and R_i^2 the coefficient of determination for a model estimating X_i from the other predictors. The *variance inflation factor* of X_i is

$$VIF_i = \frac{1}{1 - R_i^2}$$

- + Higher VIF means more multicollinearity
- + Rule of thumb: worried when $VIF_i > 5$ (i.e. $R_i^2 > 0.8$)

Class activity

Work on the class activity. Submit your rendered HTML file at the end of class.

Class activity

$$+ \widehat{\text{Hgt97}} = 40.59 + 1.10 \text{ Hgt96} \quad R^2_{adj} =$$

Class activity

$$+ \widehat{\text{Hgt97}} = 40.59 + 1.10 \text{ Hgt96} \quad R_{adj}^2 = 0.949$$

$$+ \widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96} \quad R_{adj}^2 =$$

Class activity

+ $\widehat{\text{Hgt97}} = 40.59 + 1.10 \text{ Hgt96} \quad R^2_{adj} = 0.949$

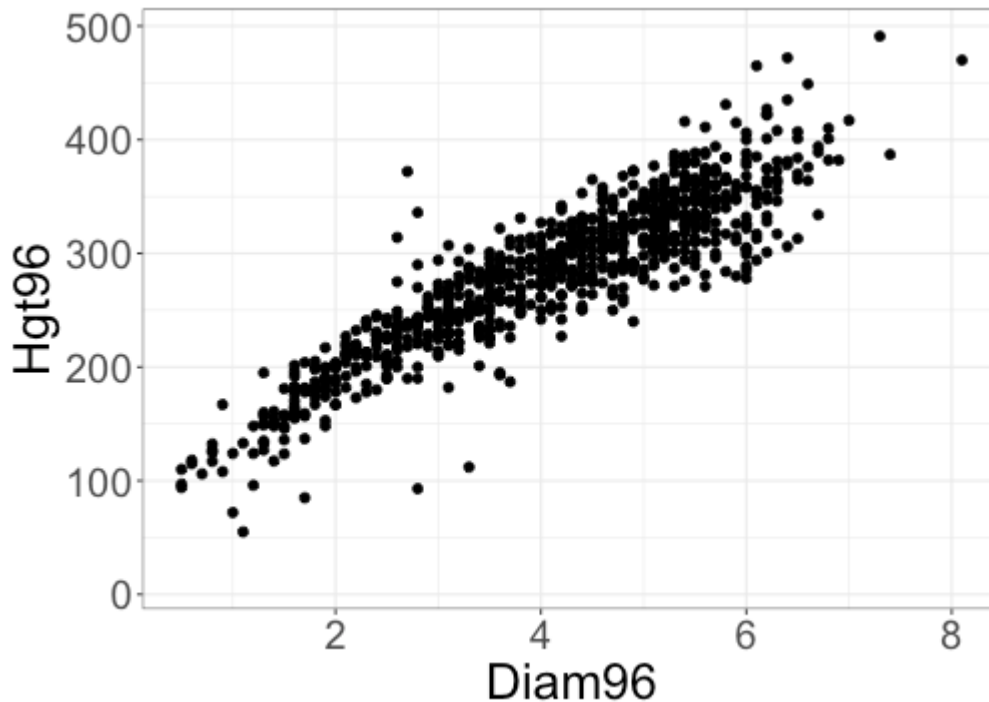
+

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96} \quad R^2_{adj} = 0.942$$

Why doesn't R^2_{adj} increase when I add diameter to the model?

Class activity

Why doesn't R^2_{adj} increase when I add diameter to the model?



Because diameter and height in 1996 are so correlated.

Class activity

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96}$$

What is the variance inflation factor for Diam96?

Class activity

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96}$$

What is the variance inflation factor for Diam96?

```
m1 <- lm(Diam96 ~ Hgt96, data = Pines)
summary(m1)$r.squared
```

```
## [1] 0.8134518
```

$$VIF = \frac{1}{1 - 0.813} = 5.35$$

Why is multicollinearity a problem?

- + Inflates variability of estimated coefficients
- + Interpretation of individual terms is difficult

$$\widehat{\sqrt{\text{MDs}}} = 3.58 + 2.58 \text{ Hospitals} + 0.012 \text{ Beds}$$

Usual interpretation: Holding Beds fixed, an increase of 1 hospital is associated with an increase of 2.58 units in $\sqrt{\text{MDs}}$

Problem: When Beds and Hospitals are highly correlated, doesn't make sense to fix one and change the other. (Can't really change the number of hospitals without changing the number of beds)